DATA ANALYSIS USING SQL

Project Report

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Course: Bachelor of Computer Applications (BCA)
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University: IFTM University

Certificate

This is to certify that Jogindra Pal, a student of BCA 5th Semester at IFTM University, has successfully completed the project titled "Data Analysis Using SQL" as part of the academic requirements for the subject Data Analysis.

Acknowledgement

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1. Introduction

SQL (Structured Query Language) is used to store, retrieve, manage, and analyze data in relational databases. This project demonstrates how SQL can be used for data analysis by creating a Student Management database and performing different analytical queries.

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2. Objectives

- Create a relational database.
- Store student information.
- Perform data analysis using SQL.
- Apply SQL functions and JOIN operations.
- Generate meaningful reports from stored data.

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3. Software Requirements

- MySQL 8.0
- MySQL Workbench
- Windows 10/11

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4. Database Design

Students

- StudentID
- Name
- Gender
- Age
- Course
- City

Subjects

- SubjectID
- SubjectName

Marks

- MarkID
- StudentID
- SubjectID
- Marks

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5. SQL Code

```
CREATE DATABASE StudentManagement;
USE StudentManagement;

CREATE TABLE Students(
StudentID INT PRIMARY KEY AUTO_INCREMENT,
Name VARCHAR(100),
Gender VARCHAR(10),
Age INT,
Course VARCHAR(50),
City VARCHAR(50)
);
```

```sql
CREATE TABLE Subjects(
SubjectID INT PRIMARY KEY AUTO_INCREMENT,
SubjectName VARCHAR(100)
);

CREATE TABLE Marks(
MarkID INT PRIMARY KEY AUTO_INCREMENT,
StudentID INT,
SubjectID INT,
Marks INT,
FOREIGN KEY(StudentID) REFERENCES Students(StudentID),
FOREIGN KEY(SubjectID) REFERENCES Subjects(SubjectID)
);
```

Sample Data

```sql
INSERT INTO Students(Name,Gender,Age,Course,City)
VALUES
('Rahul','Male',20,'BCA','Lucknow'),
('Priya','Female',21,'BCA','Kanpur'),
('Aman','Male',19,'BSc','Delhi'),
('Neha','Female',22,'BCom','Noida'),
('Rohit','Male',20,'BCA','Agra');
```

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6. SQL Queries

- Display all students.
- Display BCA students.
- Students scoring above 80 marks.
- Student-wise marks report.
- Total marks of each student.
- Average marks.
- Highest marks.
- Lowest marks.
- Count students city-wise.
- Sort students alphabetically.

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7. Results

The database successfully stores student information and allows data analysis using SQL queries. Aggregate functions and JOIN operations generate useful academic reports.

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8. Conclusion

The project demonstrates how SQL can be used for storing, managing, and analyzing educational data. It improves understanding of database design, SQL queries, and reporting techniques.

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9. References

- MySQL Documentation
- SQL Tutorial
- Database Management System Notes
- Classroom Study Material